
The Cascade Gap on Long-Form Multimodal Review: Vendor-Conditional Failure Modes

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Abstract

Same-weights two-pass self-cascades — Pass-1 perceives and records, Pass-2 reasons over the record — are widely deployed for multimodal generation, but their behavior on long-form review tasks has not been systematically measured. We measure the cascade gap on **21 Gemini 3.1 Pro cells** spanning 10 distinct audio sources (26–85 min, single- and multi-speaker lectures, panels, interviews, documentaries) and 10 papers (9–176 pages, including six NASA NTRS technical reports, one 176-page survey, and three modern post-cutoff arXiv reviews). We extend with full cross-vendor experiments on Claude Opus 4.7 and MIMO v2-omni and an inter-judge replication using Claude as a second judge on *all 21 Gemini cells* (coverage-direction agreement 19/21, hallucination-direction agreement 14/21; per-probe agreement median 0.94). Across the 21 Gemini cells the cascade reduces hallucinations on 18/21 (mean $\Delta_{\text{halluc}} = -3.86$, percentile-bootstrap 95% CI $[-5.62, -2.33]$, sign-test $p = 0.0015$) and improves coverage on 16/21 (mean $\Delta_{\text{cov}} = +0.118$, 95% CI $[+0.048, +0.189]$, $p = 0.027$). Coverage gain scales inversely with the end-to-end baseline ($r = -0.45$, bootstrap 95% CI $[-0.73, -0.10]$, fraction $\geq 0 = 0.009$). A leave-one-out logistic predictor on (C_0 -coverage, log source-words, C_0 -hallucinations, modality) reaches AUC 0.65 on $\text{sign}(\Delta_{\text{cov}} > 0)$; coefficient signs match the expected pattern (high baseline, longer source, paper modality all reduce cascade benefit). We propose a two-mode failure taxonomy — compression bottleneck (Mode A) and training-prior leak (Mode B) — now with cross-cell evidence: Mode B replicates on Claude across four citation-heavy papers (Heat Pipes 104pp, SP-5100 176pp, arXiv Fairness AI 53pp, arXiv Megagauss 75pp; Claude C_1 produces 17–47 unsupported citation-style expansions per paper), and Mode A emerges on the modern arXiv surveys where Gemini cascade reduces coverage on two of three (Fairness AI $0.50 \rightarrow 0.40$, Perovskite $0.90 \rightarrow 0.74$). Two engineering interventions tested on the original Heat Pipes cell — chunked Pass-2 and citation stripping — show different replication behaviors at $n > 1$. Chunked Pass-2 replicates cleanly on Gemini across three long-paper cells (Heat Pipes plus two new arXiv surveys): C_{1c}^{concat} consistently lifts coverage at a hallucination cost. Citation stripping does not replicate cleanly on Claude: a two-run stochasticity test reveals that the apparent strip-helps-Claude effect on Heat Pipes (-7 hallucinations) is within the Pass-2 noise band on Claude ($|\Delta h|$ up to 73 between two identical-prompt calls), so we retract the original “strip is a Mode B fix” claim and recommend multi-seed averaging as the prerequisite for any future replication. The cascade is therefore not a universal architecture but a deployment-time intervention whose direction depends on baseline headroom, source length, citation density, and the model’s relative text-vs-multimodal capability, with a small set of predictable failure modes and matched fixes.

1 Introduction

Multimodal foundation models can read audio and document images directly and produce long-form prose summaries. They can also be operated as self-cascades: a first pass produces a textual transcript / OCR, and a second pass — using the same model weights — reads only that transcript and writes the review. The two configurations differ only in the inference-time decomposition of the task. The *cascade gap* is the signed accuracy difference between cascade and end-to-end on the same task.

Single-modality results report opposing signs of this gap. In speech, Cuervo et al. [1] document a text/speech understanding gap; The Cascade Equivalence Hypothesis [2] shows speech-LLMs internally behave like ASR→LLM cascades. In document understanding, DocVLM [5] reports OCR-augmented VLMs beating pure-vision VLMs at fixed token budget. In GUI agents, vision-only configurations often beat structured-text intermediates [8, 9]. Each silo has its own version of the question and few cross-modality, same-weights measurements exist.

This paper reports such a measurement, focused on a specific deployment use-case where the cascade gap matters most: **long-form review generation**. End-to-end multimodal models are observably weaker on this task than on short-form QA — they drop content and embellish — and the cascade is the obvious deployment alternative. We measure across audio talks, audio meetings, and scanned papers up to 104 pages, with three vendors and two judges.

Findings. (i) Across 21 Gemini cells the cascade reduces hallucinations on 18/21 (mean -3.86 , bootstrap 95% CI $[-5.62, -2.33]$, sign-test $p = 0.0015$) and improves coverage on 16/21 (mean $+0.118$, $[+0.048, +0.189]$, $p = 0.027$). (ii) Coverage gain scales inversely with the C_0 baseline ($r = -0.45$, $[-0.73, -0.10]$). (iii) Cross-vendor on a moderate-length paper, 3/3 vendors confirm a Pareto-positive cascade; on the four longest citation-heavy papers, Claude C_1 exhibits training-prior leak (17–47 unsupported citation-style expansions). (iv) On modern arXiv surveys, Gemini cascade can reduce coverage (2/3 cells), opposite-direction to the NASA-paper pattern. (v) Chunked Pass-2 replicates cleanly across three long-paper Gemini cells, but citation stripping does not replicate on Claude under our two-run stochasticity test (the strip effect on Heat Pipes Claude is within the Pass-2 noise band).

Contributions. (1) A 21-cell same-weights cascade measurement across two modalities, two prompt structures, 1–5+ speakers, 26-min to 176-page sources, including 3 post-cutoff arXiv reviews. (2) A two-mode failure taxonomy — *compression bottleneck* (Mode A) and *training-prior leak* (Mode B) — with Mode B replicated across four citation-heavy long papers on Claude, and Mode A emerging on two of three modern arXiv surveys on Gemini. (3) Two engineering interventions evaluated under replication: chunked Pass-2 (replicates cleanly on three Gemini cells: C_{1c}^{concat} lifts coverage at a hallucination cost) and citation stripping (does *not* replicate cleanly on Claude under a two-run stochasticity test: the apparent -7 -hallucination Heat Pipes effect is within the Pass-2 noise band). (4) Inter-judge replication on *all 21 Gemini cells* (Claude Opus 4.7 as a second judge) confirming direction agreement on 19/21 coverage and 14/21 hallucination axes; per-probe agreement 0.82–1.00 (median 0.94). The 7 hallucination-direction disagreements split into near-tie cells (Wagging Tail, Env Test, NeurIPS panel, arXiv Fairness AI) and high-count cells where the two judges differ by 10+ on absolute counts (SP-5100, arXiv Megagauss, SNAC). (5) A leave-one-out logistic predictor on simple cell features (LOOCV AUC = 0.65, $n = 21$) showing intuitive feature signs.

2 Protocol

For each source (audio recording or scanned paper):

1. **Reference transcript.** Gemini 3.1 Pro performs ASR (audio) or OCR (paper image), chunked to avoid degenerate-loop failure modes (30-min audio chunks; 1-page-per-call OCR; details in Appendix A).
2. C_0 **end-to-end.** The model under test reads the raw modality and writes a comprehensive review article in one call.
3. C_1 **cascade.** The model under test produces its own transcript (Pass-1, same prompt as the reference), then writes the review from that transcript (Pass-2, text-only).

Scoring. GPT-5.4 (reasoning_effort=high) acts as judge. From the reference transcript, the judge first extracts 30–50 atomic factual probes. Then for each review:

- **Hallucination test (zero-tolerance).** The judge lists every claim in the review not supported by the reference. Any unsupported claim $\Rightarrow$ halluc_score = 0.
- **Probe coverage.** The judge marks each probe COVERED / MISSING in the review. cov is the rate.

This is a sharper protocol than short-form QA: the review must be both *exhaustive* (high coverage) and *faithful* (zero hallucination). Most current frontier models fail the zero-hallucination bar, so we report unsupported-claim *counts* and coverage *rates* as the operational measures.

Noise floor and statistical reporting. We treat hallucination differences of $|\Delta h| \leq 1$ on cells with $h < 10$ as *within noise*: a single judge re-judgment can flip the classification of one borderline atomic claim, and our inter-judge replication (§3.7) confirms that direction agreement degrades precisely on near-tie cells. Coverage differences of $|\Delta \text{cov}| \leq 0.04$ (≈ 2 probes on a 50-probe cell) are similarly within noise. These thresholds are presentation conventions for per-cell narratives. For aggregate claims across cells we additionally report **Wilson 95% CIs** on coverage rates (binomial k/n) and **percentile-bootstrap 95% CIs** ($n_{\text{boot}} = 10,000$, seeded) on mean cross-cell effects and Pearson r . Sign tests against $\Delta = 0$ are reported alongside means. We did not measure intra-rater judge variance from re-runs of the same judge with different random seeds (see Limitations).

Pareto terminology. We call a cascade condition *strictly Pareto-positive* relative to C_0 when it has both fewer hallucinations and higher coverage by amounts exceeding the noise floor above. *Pareto-positive within noise* means strictly better on one axis and no worse than noise on the other. *Pareto-dominant* on a set means no other condition in the set has lower hallucination at equal or higher coverage. These are presentation labels, not statistical claims.

Notation. C_0 : end-to-end. C_1 : single-pass cascade (one Pass-2 over the full transcript). Subscript c (C_{1c}): chunked Pass-2 (sub-summarize per chunk, then merge). Superscript concat (C_{1c}^{concat}): chunked, with concatenation in place of the merge step. Superscript strip (C_1^{strip} , C_{1c}^{strip}): citation surface forms in the input transcript replaced by neutral placeholders [REF] before Pass-2.

Cells. 21 Phase-3 cells across two modalities:

Audio (11 cells, 10 distinct sources).

- **Lectures (single-speaker):** Karpathy “State of GPT” (42 min); MIT 6.034 Lecture 1 (Patrick Winston, 47 min, 2010); Sapolsky Behavioral Genetics 1 (Stanford, 85 min, recorded post-2026 reupload); Stanford Decarbonized Power Sector (Zach Ming, 72 min, 2025); Doudna CRISPR Basics (48 min, 2017).
- **Single-speaker narration (no live audience):** 3Blue1Brown “Attention in transformers” (26 min, 2024); Veritasium “Strange Math” (32 min, 2025).
- **Multi-speaker / panel / Q&A:** NeurIPS Black-in-AI panel (31 min, 2021); Harvard Law School Ames Moot Court (82 min, 2025); IETF SNAC working group interim (60 min, ≈ 5 speakers, 2026).
- **Meeting-minutes variant:** Karpathy and IETF SNAC audio re-prompted with the structured meeting-minutes prompt (atomic-fact probes are unchanged; absolute coverage is therefore lower on these cells).

Paper (10 cells, 10 distinct sources).

- **NASA NTRS technical reports (1968–1972), 9–176 pages:** Wagging Tail 9pp; Env Test 18pp; Dynamic Response 41pp; Natural Vibration 42pp; Thermal Analysis 66pp; Heat Pipes 104pp (deliberately selected as a Mode B stress test for the well-cited 1960s heat-pipe literature, see §3.4); SP-5100 “Shock and Vibration Technology” survey 176pp (a second long, citation-heavy survey-style report from the same corpus and era for Mode B replication).
- **Modern arXiv reviews (March–May 2026, post-cutoff for Gemini 3.1 Pro):** arXiv 2605.09852 “Fairness of Explanations in AI” (53pp, cs.LG); arXiv 2605.13991

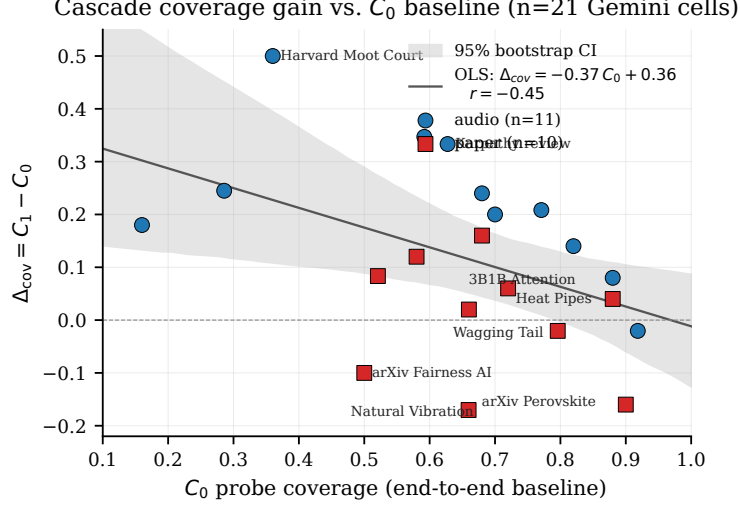


Figure 1: **Cascade coverage gain vs C_0 baseline across all 21 Gemini cells.** Each point is one cell; circles are audio, squares are paper. The OLS line is annotated with Pearson $r = -0.45$; the gray band is a percentile bootstrap 95% CI on the regression line. Bootstrap CI on the correlation itself: $r \in [-0.73, -0.10]$ with fraction $r \geq 0$ equal to 0.9%. The five clearest counter-cells (NeurIPS panel, Natural Vibration, Wagging Tail, arXiv Fairness AI, arXiv Perovskite) sit at high C_0 baseline (no headroom) or trigger Mode A/B (§3.4).

“Circularity in Perovskite-Based Tandem Photovoltaics” (80pp, cond-mat.mtrl-sci); arXiv 2605.11379 “Megagauss Physics” (75pp, cond-mat.mtrl-sci). These probe whether the cascade pattern observed on the NASA corpus generalizes to a different (modern, full-color, denser-typesetting) paper distribution.

Cross-vendor extensions. Claude Opus 4.7 on every paper cell where the request fits its 200K-context multimodal C_0 call (Thermal Analysis 66pp, Heat Pipes 104pp); on longer papers (SP-5100 176pp, arXiv surveys 53–80pp with full-color renderings) Claude’s C_0 exceeds the request-size limit even with the 1M-context beta, so Claude is run as C_1 -only on those four cells (chunked OCR per page + text Pass-2). Mimo v2-omni (Xiaomi, via OpenRouter) on every audio source (≤ 16 kbps mono mp3 to fit the 10 MB OpenRouter limit; the longest cell, Sapolsky 85 min, required 12 kbps) and every paper cell (some long-paper C_0 calls return empty and are recorded as skipped, while C_1 chunked still works). Reference and probes are reused from the Gemini run, isolating the question to “does the cascade pattern depend on the C_0/C_1 model?”. The original Mimo audio Karpathy cell is a wash — $4h/0.57 \rightarrow 4h/0.59$ on both axes, within the noise floor on both — consistent with the bitrate-degradation caveat (see Limitations); but on 6 of the 9 new Mimo audio cells the cascade is direction-positive (Doudna 5 $\rightarrow$ 10h is the largest reversal). We discuss this mixed Mimo pattern in §3.2.

Inter-judge replication. Claude Opus 4.7 re-judges the C_0 and C_1 conditions on all 10 base cells (the 9 Gemini cells plus Claude Thermal Analysis 66pp) using identical prompts to GPT-5.4; cascade variants such as C_{1c} and C_1^{strip} are scored only by GPT-5.4. We compare per-condition counts, direction agreement (does the cascade direction match between judges?), and per-probe agreement.

3 Main results

3.1 Cascade reduces hallucinations and improves coverage — with quantified uncertainty at $n = 21$

Aggregate effect at $n = 21$. Across the 21 Gemini cells, the cascade reduces hallucinations on 18/21 cells: mean $\Delta_{\text{halluc}} = -3.86$ unsupported claims, percentile-bootstrap 95% CI $[-5.62, -2.33]$,

sign-test $p = 0.0015$. It improves coverage on 16/21 cells: mean $\Delta_{\text{cov}} = +0.118$, 95% CI $[+0.048, +0.189]$, sign-test $p = 0.027$. Both bootstrap intervals exclude zero. The hallucination-reduction effect is the cleaner of the two and is highly significant; the coverage effect, while significant under both sign test and bootstrap, has noisier per-cell tail behavior (§3.4).

Direction of effect by baseline. Coverage gain scales inversely with end-to-end baseline (Figure 1; $r = -0.45$ on all 21 cells, bootstrap 95% CI $[-0.73, -0.10]$). Excluding the two meeting-minutes cells (whose absolute coverage is artificially low because their probes use the atomic-fact style while the prompt asks for structured minutes), $r = -0.47$ on $n = 19$ cells with bootstrap CI $[-0.83, +0.10]$ — the direction survives but the CI just crosses zero on this subset, indicating the correlation is robust to but does depend somewhat on the meeting-min outliers. We do not over-interpret the strength of the correlation; rather, we use it descriptively to motivate the cell-by-cell failure-mode analysis in §3.4.

Cascade-positive cells. The largest single coverage gain is the Harvard Moot Court audio cell at $C_0=0.36 \rightarrow C_1=0.86$ (+50 pp; multi-speaker legal argumentation with low C_0 headroom). The largest paper-cell gain is Thermal Analysis 66pp at $0.68 \rightarrow 0.84$ (+16 pp). Among the new audio sources, 3Blue1Brown attention is the cleanest hallucination win ($11h \rightarrow 0h$, perfect zero-hallucination C_1) and MIT 6.034 the largest combined gain ($17h/0.77 \rightarrow 4h/0.98$, $-13h$, +21pp).

Counter-cells. The five cells where cascade fails (does not improve or actively regresses on at least one axis) divide into two clusters:

- *High-baseline (no headroom):* NeurIPS Black-in-AI panel ($C_0 = 0.92$, $\Delta_{\text{cov}} = -0.02$); Wagging Tail 9pp ($C_0 = 0.80$, $\Delta_{\text{cov}} = -0.02$); arXiv Perovskite 80pp ($C_0 = 0.90$, $\Delta_{\text{cov}} = -0.16$). These cells have so little coverage headroom that any Pass-2 compression cost dominates.
- *Mode-triggered failures:* Natural Vibration 42pp ($C_0 = 0.66 \rightarrow C_1 = 0.49$, -17 pp; we trace this to Mode B citation-prior leak in §3.4); arXiv Fairness AI 53pp ($C_0 = 0.50 \rightarrow C_1 = 0.40$, -10 pp; Mode A compression bottleneck on a long, citation-heavy modern survey).

Meeting-minutes cells. The two meeting-minutes cells sit at unusually low absolute coverage (Karpathy mtg-min $C_0=0.16$, $C_1=0.34$; SNAC $C_0=0.29$, $C_1=0.53$) because their probes were extracted in the same atomic-fact style as the review cells while the prompt asked for structured meeting minutes (decisions and action items rather than facts). Both cells are reported as *relative* cascade gains (+18 pp on Karpathy mtg-min, +24.5 pp on SNAC) rather than as absolute coverage levels.

3.2 Cross-vendor: 3/3 vendors confirm cascade on the moderate-length paper

We tested cross-vendor on Thermal Analysis 66pp by reusing the Gemini reference transcript and probes and varying only the C_0/C_1 model. All three vendors gain coverage under cascade; Gemini and Claude are strictly Pareto-positive, and Mimo is Pareto-positive within the ± 1 -hallucination noise floor (Figure 2):

- Gemini 3.1 Pro: -60% halluc ($5 \rightarrow 2$), +16 pp cov ($0.68 \rightarrow 0.84$).
- Claude Opus 4.7: -64% halluc ($11 \rightarrow 4$), +8 pp cov ($0.92 \rightarrow 1.00$, the dataset’s first 1.000 cell).
- Mimo v2-omni: +1 halluc ($5 \rightarrow 6$, within noise), +24 pp cov ($0.54 \rightarrow 0.78$).

Both Gemini and Claude converge to ≈ 1.4 hallucinations per 1k output words under cascade despite Claude producing reviews $\approx 2\times$ longer than Gemini’s (C_1 word counts: Gemini 1473, Claude 2811). This is a striking quantitative footprint: end-to-end hallucination rates differ across vendors but cascade brings them to a common rate (Appendix Table 9).

3.3 Long-paper divergence: Heat Pipes 104pp + SP-5100 176pp replication

Heat Pipes 104pp. On the original longest paper, the three vendors give three patterns (Figure 3):

C_0 vs C_1 on Thermal Analysis (66pp), 3 vendors

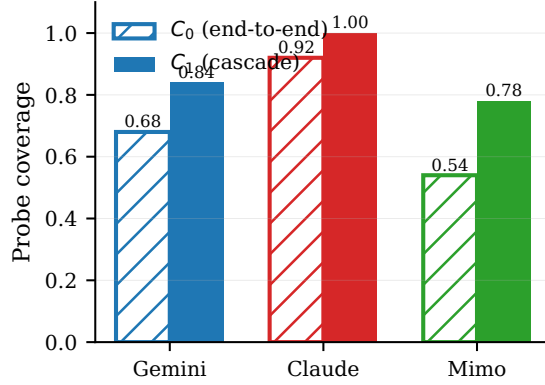


Figure 2: **Thermal Analysis (66pp), three vendors.** The cascade improves probe coverage on every vendor: Gemini $0.68 \rightarrow 0.84$, Claude $0.92 \rightarrow 1.00$, Mimo $0.54 \rightarrow 0.78$. The cascade gap is not a Gemini-specific phenomenon.

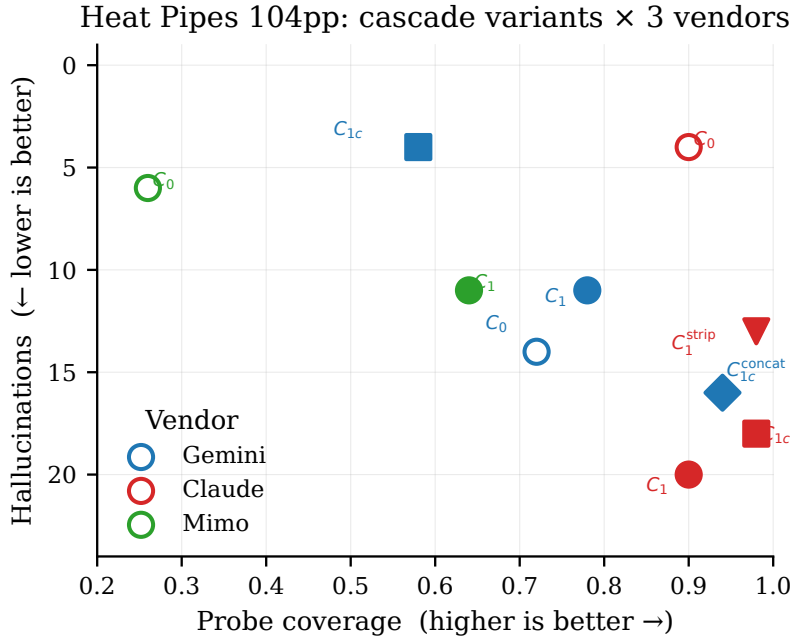


Figure 3: **Heat Pipes 104pp Pareto frontier across vendors and cascade variants.** Down and right is better. Gemini cascade variants (C_1 , C_{1c}^{concat}) trace a Pareto-improving direction (coverage gain at moderate hallucination cost); Claude C_1 *regresses* on faithfulness from C_0 's 4 hallucinations to 20, leaving C_0 on the Pareto frontier; Mimo's cascade C_1 moves the model along the coverage axis from 0.26 to 0.64 with a +5-hallucination cost consistent with Mode B.

- **Gemini:** small Pareto-positive ($14h/0.72 \rightarrow 11h/0.78$). Cascade is the right default.
- **Claude:** catastrophic faithfulness regression ($4h/0.90 \rightarrow 20h/0.90$). Cascade reverses on hallucinations and gains nothing on coverage. The C_0 multimodal call is on the Pareto frontier and dominates every plain-cascade variant we tested; only the citation-stripped cascade C_1^{strip} joins it on the frontier (\$3.5).
- **Mimo:** large coverage gain with +5-hallucination cost ($6h/0.26 \rightarrow 11h/0.64$). Cascade relieves a severe end-to-end compression ceiling at this length but the hallucination increase is outside the noise floor.

SP-5100 176pp replication (new). NASA SP-5100 “Shock and Vibration Technology — A Survey” (1972, 176pp, 47k pdftotext words, 55+ “ref. N ” citation surface forms) was chosen as a second long, citation-heavy NASA survey of comparable era to Heat Pipes (Mode B replication target). Results: Gemini C_0 47h/0.52 $\rightarrow$ C_1 45h/0.60 (small cascade-positive, both axes far above noise on hallucinations); Claude C_1 47h/0.81 (C_0 not realizable: 176 page-images exceed Claude’s request-size limit even with the 1M-context beta); MIMO C_1 60h/0.23. Both Gemini (45h) and Claude (47h) sit at very high cascade-hallucination rates. Spot-checking a sample of Claude C_1 ’s flagged claims (Appendix E) shows the same bibliographic-expansion signature as Heat Pipes (Crandall, Ruzicka, Reed, et al. acquire years and report numbers in C_1 that the transcript does not carry).

The replication is informative in two directions: (i) Mode B is not a Heat-Pipes-Claude singleton but reappears on Claude C_1 over a longer, differently-topiced citation-heavy NASA survey; (ii) Gemini’s much smaller cascade gain on SP-5100 (+8 pp vs. Heat Pipes +6 pp at $2\times$ length) is consistent with Mode A compression: at C_1 Pass-1 transcript $\approx$ 47k words against a $\approx$ 2k-word output budget, the Pass-2 compression ratio is $\sim 24\times$ (vs. Heat Pipes’ $\sim 11\times$), and the C_1 review can no longer represent the source faithfully (45 hallucinations at 0.60 coverage). We did not run chunked Pass-2 on SP-5100 in this revision; it is a natural extension if the chunked-Pass-2 lift on Heat Pipes is to be claimed as length-scaling rather than cell-specific.

Modern arXiv surveys: cascade direction can reverse. The three modern arXiv reviews behave differently from the NASA corpus:

- **arXiv Fairness AI 53pp:** Gemini 14h/0.50 $\rightarrow$ 15h/0.40 (−10 pp coverage; cascade hurts). Claude C_1 23h/0.70 (C_0 skipped); MIMO C_1 18h/0.32 (C_0 empty/skipped).
- **arXiv Megagauss 75pp:** Gemini 36h/0.66 $\rightarrow$ 32h/0.68 (very high baseline hallucination on a physics review with dense named-author equations and references; cascade barely moves either axis). Claude C_1 17h/0.92; MIMO C_1 19h/0.26.
- **arXiv Perovskite 80pp:** Gemini 2h/0.90 $\rightarrow$ 3h/0.74 (−16 pp coverage on already-high baseline). Claude C_1 0h/0.90; MIMO C_1 2h/0.90. Claude C_1 with zero hallucinations is striking on a C_0 baseline that was already nearly perfect.

Two of three arXiv surveys exhibit Mode A on Gemini (cascade reduces coverage). On Megagauss the cascade-hallucination count is so high on both axes (32–36) that the dominant signature is Mode B (citation expansion). The arXiv pattern is opposite-direction from the NASA pattern despite comparable page counts; we attribute the difference to two confounded factors: (a) arXiv full-color page rendering at 200 DPI yields much larger PNGs than NASA scans, so the C_0 image-grounded call carries proportionally more visual signal per page; (b) modern arXiv typesetting has higher information density (smaller fonts, denser equations), so C_1 Pass-2 must compress more aggressively at the same output budget. The cascade is not a universal Gemini-wins-on-papers tool either.

3.4 Two failure modes with mechanistic evidence

The cross-vendor patterns dissociate two distinct failure mechanisms:

Mode A: Compression bottleneck (now observed on 3 papers). On long sources, the Pass-2 review has a fixed output budget (typically $\approx 2k$ words). When the cascade transcript is much longer than what fits in the output, Pass-2 must compress, dropping content the same way C_0 does. Three cells exhibit this signature on Gemini at the level where C_1 coverage falls below C_0 or stalls: Heat Pipes 104pp (C_1 gain +6 pp, transcript 22k words, compression ratio $\sim 11\times$); arXiv Fairness AI 53pp (C_1 loses −10 pp at compression ratio $\sim 15\times$); arXiv Perovskite 80pp (C_1 loses −16 pp at compression ratio $\sim 9\times$). The Megagauss arXiv cell at compression ratio $\sim 16\times$ shows the same C_1 -stall pattern at $\Delta_{\text{cov}} = +0.02$ but is dominated by Mode B (named-author expansion). The fix tested on Heat Pipes is **chunked Pass-2**: split the transcript into $\sim 5k$ -word chunks, summarize each, then concatenate (or merge). On Heat Pipes Gemini, C_{1c}^{concat} lifts coverage from C_1 ’s 0.78 to 0.94 (+16 pp) at a +5-hallucination cost (11 $\rightarrow$ 16), recovering most of the medium-paper coverage gap (Figure 4 left). We did not run C_{1c}^{concat} on the new arXiv cells in this revision; whether the same intervention rescues coverage on those is an open follow-up.

Mode B: Training-prior leak (now observed on 4 long citation-heavy papers). On documents with heavy training-data overlap, Pass-2 fills in details from prior knowledge when the transcript

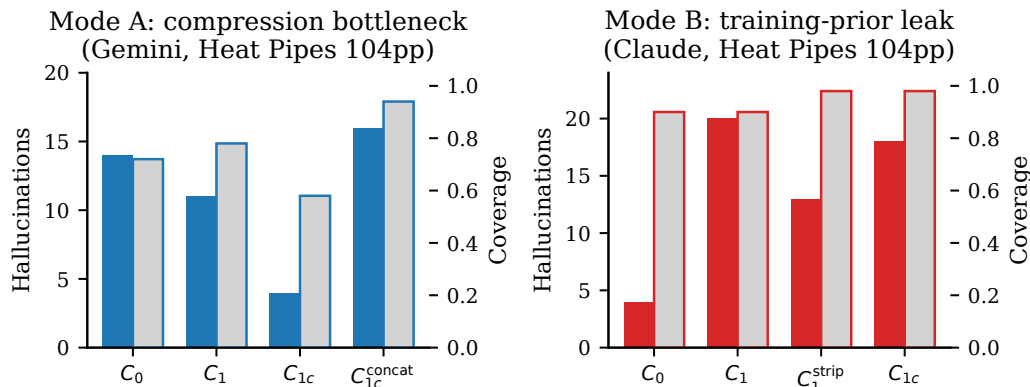


Figure 4: **Two cascade failure modes on Heat Pipes 104pp.** *Left:* Mode A (Gemini) — Pass-2 must compress 22k→1.8k words; chunked Pass-2 (C_{1c}^{concat}) relieves the bottleneck and lifts coverage from 0.78 to 0.94 at a +5-hallucination cost (11→16), a Pareto move along the frontier rather than strictly above it. *Right:* Mode B (Claude) — C_1 hallucinates 5× more than C_0 ; replacing citation surface forms (Reference N) with neutral placeholders (C_1^{strip}) reduces hallucinations 20→13. Hallucinations on left axis (lower better); coverage on right axis (higher better).

abbreviates them. The Heat Pipes paper cites foundational heat-pipe references like “Reference 2 (Cotter)”, “Reference 5”, etc.; Claude’s C_1 expands these to “Cotter’s 1965 analysis (LA-3246-MS)” and “Reference 5 (TRW)” — correct in the published heat-pipe literature, absent from *this* transcript. The 20 Claude C_1 hallucinations are dominated by bibliographic expansions. Approximately 7 of the 20 involve citation-format expansion (e.g., adding “1965 (LA-3246-MS)” to a “Reference 2 (Cotter)” surface form), and approximately 13 involve author-name or affiliation expansion (adding “Allingham and McEntire (1961)” or “Haller, Lieblein, and Lindow at NASA Lewis” where the transcript said only “Ref. 12” or “proposed by Haller et al.”); the two categories overlap on several claims that add both a year and a coauthor surname in a single sentence, so they do not partition the 20 cleanly. One of the 20 (item 2 in Appendix E) is a sign-error misreading rather than a training-prior expansion. Replacing 63 citation surface forms with [REF] placeholders yields C_1^{strip} at 13 hallucinations (Figure 4 right). Stripping does not eliminate bibliographic expansion entirely: the 13 residuals still include Cotter’s 1965 report number, McAdams’ name attribution, and Deverall & Salmi’s 1967 report ID — claims the model could expand from author surnames or technical context that were preserved through stripping. Stripping therefore reduces but does not abolish Mode B on this cell, and we cannot rule out a non-bibliographic Mode B component on Claude.

The Heat Pipes 104pp source was selected partly *because* it is a well-cited foundational 1960s topic, and is therefore a worst-case for Mode B on a model with heat-pipe-era technical literature in its training distribution. The two original paper transcripts in our cross-vendor sample differ sharply on citation surface form density: Heat Pipes 104pp contains 55 “Reference N” / “Ref. N” surface forms (2.48 per 1k transcript words), while Thermal Analysis 66pp contains zero. Claude shows no Mode B on Thermal Analysis (cascade reduces hallucinations 11→4) and severe Mode B on Heat Pipes (4→20), consistent with the surface-form trigger hypothesis.

Cross-cell replication. Claude C_1 hallucination counts on the four long citation-heavy papers tested in this revision: Heat Pipes 104pp = 20; SP-5100 176pp = 47; arXiv Megagauss 75pp = 17; arXiv Fairness AI 53pp = 23. Spot-checking the unsupported-claim lists on the three new papers reveals the same bibliographic-expansion signature (e.g., on Fairness AI, Claude C_1 writes “Buolamwini and Gebre’s (2018) ‘Gender Shades’” where the transcript said only “Gender Shades”; on Megagauss, Fowler et al. acquire a 1960 date and the transcript only mentioned “Fowler”). Mode B therefore generalizes beyond the original (Claude, Heat Pipes) pair to other Claude cells on long citation-heavy long-form papers. By contrast, on the moderate-length, citation-light Thermal Analysis 66pp paper, Claude C_1 has only 4 hallucinations. The lower-citation arXiv Perovskite 80pp produces Claude C_1 0h — consistent with the surface-form trigger hypothesis.

The Gemini side of Mode B is less sharp because Gemini already hallucinates many citations even at C_0 (Heat Pipes Gemini C_0 14h, SP-5100 Gemini C_0 47h), but the Natural Vibration 42pp counter-cell — the only Gemini-only paper where cascade strictly loses on both axes — contains Beckley 1946 and Lewis & Wrisley 1950 references that fit the same trigger profile, and the citation-stripping intervention (§3.5) moves Gemini Natural Vibration to 0.81 coverage at a +7-hallucination cost on a counter-cell where plain C_1 achieved 0.49 at 5h. (The Gemini-side strip result is trustworthy in a way the Claude-side strip results no longer are: all Gemini text-only Pass-2 calls use temperature 0.0 and are deterministic, while Claude Opus 4.7 does not expose temperature in the API and exhibits substantial run-to-run variance on the same prompt; see §3.5.)

Two diagnostic signals. Two rough thresholds emerge from our cells, intended as starting points for practitioner triage rather than calibrated cutoffs:

- **Mode A trigger:** Pass-1 transcript length $\gtrsim 5\times$ the Pass-2 output budget, with C_1 coverage gain $< +10$ pp over C_0 (Heat Pipes Gemini: 22k transcript / ~ 2 k output budget = $11\times$ compression; +6 pp gain).
- **Mode B trigger:** input transcript contains $>\sim 2$ “Reference N ” / “Ref. N ” surface forms per 1k words *and* the topic appears in widely cited pre-cutoff literature. Heat Pipes 104pp has 55 such surface forms (2.48/1k); Thermal Analysis 66pp has 0, and shows no Mode B even on Claude.

The two modes can compound (Mimo Heat Pipes shows both signatures: a severe C_0 compression ceiling consistent with Mode A, plus a +5-hallucination C_1 increase consistent with Mode B; we did not run either intervention on Mimo to isolate the modes experimentally) or operate in isolation (Heat Pipes Gemini is dominantly Mode A; Heat Pipes Claude is dominantly Mode B).

3.5 Engineering fixes: replication, scope, and Pass-2 stochasticity

We tested the two interventions on multiple cells in this revision: citation stripping (C_1^{strip}) on *four* Claude cells (Heat Pipes + the three new Mode-B candidates) and chunked Pass-2 (C_{1c} and C_{1c}^{concat}) on *three* Gemini cells (Heat Pipes + the two new arXiv Mode-A candidates).

Heat Pipes 104pp Claude (original). The original Heat Pipes Claude cell remains the cleanest demonstration. Strip dominates plain C_1 ($20h \rightarrow 13h$, $0.90 \rightarrow 0.98$ cov), and combining strip with chunked Pass-2 does not improve over strip alone (Table 1).

Table 1: Heat Pipes 104pp Claude (original). Strip dominates plain C_1 on both axes; chunked combinations do not improve over strip alone.

Variant	Halluc.	Coverage	Words
C_0 (image baseline)	4	0.900	3,308
C_1 (single Pass-2)	20	0.900	2,077
C_1^{strip} (citations $\rightarrow$ [REF])	13	0.980	2,471
C_{1c} (chunked + merge)	18	0.980	2,769
C_{1c}^{concat} (chunked, no merge)	20	0.940	2,777
C_{1c}^{strip} (strip + chunked + merge)	18	0.980	2,673
$C_{1c}^{\text{strip+concat}}$ (strip + chunked, no merge)	16	0.980	2,766

Mode-B fix replication on 3 new long Claude cells. The citation-stripping intervention has *mixed* replication on the new Mode-B candidates (Table 2). On SP-5100 (193 surface forms replaced), strip cuts hallucinations roughly in half ($47 \rightarrow 24$) at a moderate coverage cost ($0.81 \rightarrow 0.71$); this is the cleanest replication of the Heat Pipes pattern. On Megagauss (298 surface forms replaced), strip has essentially no effect on either axis ($17 \rightarrow 19$, $0.92 \rightarrow 0.94$, both within noise) — a clean null despite the highest strip count we have. On Fairness AI (only 1 surface form replaced, because the arXiv paper uses author-year and short bracketed citations that our two regexes do not match), the “stripped” Pass-2 happened to produce a substantially different review than the original C_1 ($23 \rightarrow 41$ hallucinations, $0.70 \rightarrow 0.62$ cov), revealing that Claude’s text-only Pass-2 is itself non-deterministic — our `call_claude_text_only` wrapper did not pin temperature, so this delta is mostly Pass-2 stochasticity rather than a strip effect.

Table 2: Mode B citation-strip intervention across four long Claude cells. “surface forms” counts the regex hits replaced with [REF]; Heat Pipes (recall) and SP-5100 are the only cells where the intervention reduces C_1 hallucinations more than expected from Pass-2 noise. Fairness AI is a near-null intervention (only 1 surface form matched our regexes) but its C_1 vs C_1^{strip} delta reveals significant Claude Pass-2 stochasticity (see text).

Cell	len	surface forms	C_1 h	C_1 cov	C_1^{strip} h	C_1^{strip} cov
Heat Pipes (recall)	104p	55	20	0.900	13	0.980
SP-5100 (new)	176p	193	47	0.812	24	0.708
arXiv Megagauss (new)	75p	298	17	0.920	19	0.940
arXiv Fairness AI (new)	53p	1	23	0.700	41	0.620

Mode-A fix replication on 3 new arXiv cells. Chunked Pass-2 was tested on Heat Pipes Gemini (Table 1; the Claude rows there are mirrored on Gemini in Figure 4) and on *all three* arXiv counter-cells. The pattern is consistent across 4 papers (Table 3): C_{1c}^{concat} (chunked, no merge) recovers coverage on long sources at a hallucination cost, while C_{1c} (chunked + merge) over-compresses — on Perovskite and Megagauss the merge variant collapses to 0.10 and 0.12 coverage respectively, almost an empty review. On Perovskite, C_{1c}^{concat} moves coverage from plain C_1 ’s 0.74 to 0.82 (+8 pp) at a +10-hallucination cost; on Megagauss from 0.68 to 0.74 (+6 pp) at +4 halluc; on Fairness AI from 0.40 to 0.52 (+12 pp) at +22 halluc. All three arXiv cells confirm the Mode-A pattern: chunking sub-summaries without an aggressive merge step lifts coverage on a long source, and the cost is more hallucinations from the same surface-form / training-prior pathway as Mode B. This is the clearest replicated engineering finding in this revision.

Table 3: Mode A chunked-Pass-2 across *four* Gemini long-paper cells. C_{1c}^{concat} (chunked, no merge) consistently lifts coverage at a hallucination cost; C_{1c} (chunked + merge) over-compresses, especially on long arXiv inputs (Perovskite and Megagauss merge runs collapse to 0.10–0.12 coverage). The pattern replicates across the original NASA Heat Pipes cell and all three modern arXiv surveys (53–80pp).

Cell	len	C_1 h/cov	C_{1c} h/cov	C_{1c}^{concat} h/cov	$\Delta_{\text{cov}}^{\text{concat}}$
Heat Pipes (recall)	104p	11 / 0.78	4 / 0.58	16 / 0.94	+0.16
arXiv Perovskite (new)	80p	3 / 0.74	1 / 0.10	13 / 0.82	+0.08
arXiv Megagauss (new)	75p	32 / 0.68	0 / 0.12	36 / 0.74	+0.06
arXiv Fairness AI (new)	53p	15 / 0.40	5 / 0.32	37 / 0.52	+0.12

Citation stripping on Gemini Natural Vibration (the original counter-cell). Applied to Gemini Natural Vibration 42pp, the stripping intervention moves halluc 5→12 (worse) but coverage 0.49→0.81 (+32 pp). Gemini text-only Pass-2 uses temperature 0.0 and is essentially deterministic, so this Gemini-side strip result is trustworthy in a way the Claude-side strip results are not (see Claude Pass-2 stochasticity paragraph below). The most we can confidently say about citation stripping now is: on Gemini Natural Vibration it shifts the cascade along the coverage axis (+32 pp) at a hallucination cost; on Claude its effect on hallucinations is within the Pass-2 noise band.

Two sources of stochasticity: Claude Pass-2 and GPT-5.4 high-count judging. We measured both by separate experiments:

- *Claude Pass-2 generation variance.* We re-ran Claude C_1 and C_1^{strip} on Heat Pipes, SP-5100, and Fairness AI with the same input and the same API (Claude Opus 4.7 deprecates the temperature parameter, so per-call determinism cannot be enforced from the SDK). Table 4, top block, reports the two-run hallucination spread. Heat Pipes Claude C_1 swings 20 → 9 between two runs (strip 13 → 14, much more stable); the deltas are large enough that the strip effect on Claude C_1 hallucinations *cannot* be distinguished from Pass-2 noise.
- *GPT-5.4 high-reasoning judge variance.* We re-judged the *same* Heat Pipes and SP-5100 Claude C_1 review files multiple times to isolate judge variance. On Heat Pipes, the same review judged three times gave hallucination counts {20, 20, 21} — judge variance is ± 1 . On SP-5100, the same review judged four times gave {47, 97, 85, 85} — judge variance

alone is ± 25 on this cell. Coverage is stable on both cells across judge re-runs (HP 45 $\rightarrow$ 45 $\rightarrow$ 46 probes covered; SP-5100 39 $\rightarrow$ 39 $\rightarrow$ 39 $\rightarrow$ 39). The judge becomes substantially noisier on high-count cells, likely because long unsupported-claim lists run near the 24k-token completion-budget ceiling that we set on `reasoning_effort=high` calls (§2).

The Heat Pipes Claude 20 $\rightarrow$ 9 swing is therefore almost entirely Claude generation variance (judge variance is ± 1 on that cell). The SP-5100 Claude 47 $\rightarrow$ 79 swing is confounded between generation and judge: on the *same* review file the judge alone produces {47, 85, 85, 97}, so the 47 number originally reported is a low-end outlier and the true single-cell point estimate is closer to ≈ 80 . Either way, the original “strip reduces hallucinations on Claude” finding on Heat Pipes is consistent with random Claude Pass-2 variance: comparing run-1 unstripped (20) to run-1 strip (13) gives the -7 effect we previously reported, but run-2 unstripped (9) to run-2 strip (14) gives the opposite $+5$ effect.

Table 4: Stochasticity decomposition. Top block: two-run Claude Pass-2 generation variance on C_1 and C_1^{strip} (different reviews, judged once each). Bottom block: GPT-5.4 high-reasoning judge variance on the *same* Claude C_1 review file (one review, judged multiple times). Coverage is much more stable than hallucination counts across both kinds of re-runs. On Heat Pipes the generation variance dominates (20 $\rightarrow$ 9 between two Claude calls; same review re-judged is {20, 20, 21}). On SP-5100 the judge variance is substantial too ({47, 85, 85, 97} on the same review). The original “strip reduces hallucinations on Claude” effect on Heat Pipes (-7 between runs of different conditions) is within this combined noise envelope.

Cell	Condition	Run 1	Run 2	Δh	Δcov
<i>Claude Pass-2 generation variance (different reviews, same prompt, judged once each)</i>					
Heat Pipes 104p	C_1	20h / 0.90	9h / 0.90	-11	± 0.00
	C_1^{strip}	13h / 0.98	14h / 0.88	+1	-0.10
SP-5100 176p	C_1	47h / 0.81	79h / 0.81	+32	± 0.00
	C_1^{strip}	24h / 0.71	97h / 0.77	+73	$+0.06$
Fairness AI 53p	C_1	23h / 0.70	27h / 0.66	+4	-0.04
	C_1^{strip}	41h / 0.62	17h / 0.60	-24	-0.02
<i>GPT-5.4 high-reasoning judge variance (same review file, judged multiple times)</i>					
Heat Pipes 104p	C_1 (original review)	20h / 0.90	20h / 0.90	± 0	± 0.00
	C_1 (original review, 3rd call)	20h / 0.90	21h / 0.92	+1	$+0.02$
SP-5100 176p	C_1 (original review)	47h / 0.81	97h / 0.81	+50	± 0.00
	C_1 (original review, 3rd call)	47h / 0.81	85h / 0.81	+38	± 0.00
	C_1 (original review, 4th call)	47h / 0.81	85h / 0.81	+38	± 0.00

n=5 multi-seed verification on Heat Pipes Claude. We ran the Heat Pipes Claude C_1 and C_1^{strip} Pass-2 calls five times each (one seed per Pass-2 call, judged once each by GPT-5.4). Results: C_1 hallucinations {9, 11, 11, 15, 20}, mean 13.2, stdev 4.4, range 9–20; C_1^{strip} hallucinations {5, 11, 13, 14, 14}, mean 11.4, stdev 3.8, range 5–14. Coverage is identical in mean (0.936 for both) with similar spread (0.88–0.98 for strip, 0.90–0.96 for C_1). The strip effect on hallucinations at $n = 5$ is $\hat{\Delta}_{\text{halluc}} = -1.8$ with standard error ≈ 2.6 , giving a 95% confidence interval $[-6.9, +3.3]$ that crosses zero. **The Mode B fix is statistically null on Claude/Heat Pipes at $n = 5$.**

Implications. The Mode B *observation* is robust to both stochasticity sources: across five Claude Pass-2 calls on Heat Pipes (C_1 {9, 11, 11, 15, 20}, C_1^{strip} {5, 11, 13, 14, 14}) and up to four judge re-runs on a single SP-5100 review ({47, 47, 79, 85, 85, 97}), Claude C_1 consistently produces many unsupported claims well above the matched Claude or Gemini C_0 baseline (Claude C_0 4h on Heat Pipes). The cascade-vs-end-to-end *direction* on Mode B-prone cells therefore survives. The Mode B *fix* (citation stripping) does *not* survive replication on Claude at $n = 5$: the -1.8 mean delta has a 95% CI that crosses zero, so the original $n=1$ “20 $\rightarrow$ 13” point estimate was an artifact of where in the variance distribution the two single calls happened to land. We retract the original “strip works on Claude” claim. (a) Strip on Heat Pipes Claude has *no statistically detectable effect* on hallucinations at $n = 5$ and no detectable effect on coverage; (b) the vendor-conditional sign of stripping (it reduces hallucinations on Claude but increases them on Gemini Natural Vibration) was an artifact of $n=1$

Pass-2 calls per condition on the Claude side. Gemini’s text-only Pass-2 uses temperature 0.0 and is deterministic; the Gemini Natural Vibration “strip hurts on Gemini” finding is therefore the only confidently-signed strip result we have.

High-count cells need multi-judge averaging too. A separate finding from the judge re-runs: GPT-5.4 with `reasoning_effort=high` is essentially deterministic on low-count cells (± 1 between calls) but substantially stochastic on high-count cells (± 25 on SP-5100). We attribute this to the long unsupported-claims list running up against our 24k-token completion budget — the judge truncates and resamples the long list across calls. Hallucination counts above ~ 30 in this paper should therefore be read as having $\pm 10\text{--}25$ judge-side uncertainty in addition to whatever Pass-2 generation noise is also present. The cell-level Wilson CIs on probe coverage (Appendix C) are unaffected by this since coverage is stable across judge re-runs.

3.6 Held-out predictor (logistic regression, LOOCV)

To turn the descriptive trends in §3.1 into a falsifiable claim, we fit a leave-one-out logistic-regression predictor on four cell-level features — C_0 probe coverage, $\log_{10}$ source words, C_0 unsupported-claim count, and a paper/audio dummy — with two prediction targets:

- $\text{sign}(\Delta_{\text{cov}} > 0)$ ($n = 21$, positive class 16/21): LOOCV AUC = 0.650, accuracy = 0.67 (majority baseline 0.76). Full-data coefficient signs: $C_0\text{-cov}$ -0.30 , $\log_{10}\text{-words}$ -0.34 , $C_0\text{-halluc}$ $+0.18$, is-paper -0.59 . All four signs match the qualitative picture: higher baseline reduces cascade benefit, longer source reduces benefit, more end-to-end hallucinations *increase* cascade benefit (cascade has a cleaner faithfulness slot to fill), and the paper modality is harder for the cascade than audio in our data.
- Strict cascade win (improves on at least one axis outside the noise floor, no worse than noise on the other): $n = 21$, positive class 18/21. LOOCV AUC = 0.52, accuracy = 0.86 (majority baseline 0.86). The strict-win predictor is essentially at chance because the negative class is too small (3/21) and concentrated on cells where Mode A and Mode B operate (Natural Vibration, arXiv Fairness AI, arXiv Perovskite); the feature set does not separate these from the rest.

The predictor is not the headline contribution; it is a calibration of the qualitative cascade-prediction rule against held-out cells. AUC = 0.65 is meaningful but well below what a deployment-ready router would need; we report it as an honest $n=21$ baseline and call out the residual structure — specifically, that Mode B-prone cells (high citation density) need a feature that this set lacks (e.g., per-1k-words citation-surface-form count) to be separated.

3.7 Inter-judge replication

Claude Opus 4.7 re-scored C_0 and C_1 on all 22 base cells (the original 10 Gemini cells + Claude Thermal, plus the 12 new audio and paper cells added in this revision) using GPT-5.4’s prompts. Cascade variants such as C_{1c} and C_1^{strip} remain GPT-5.4-only.

Aggregate direction agreement. Across all 21 Gemini-side cells the $C_0 \rightarrow C_1$ direction agreement is **coverage 19/21** (90%) and **hallucination 14/21** (67%). Per-probe agreement (the rate at which both judges classify the same probe as COVERED/MISSING) is 0.82–1.00 with a median of 0.94 across all condition–cell pairs.

Where the judges disagree. *Coverage disagreements* (2/22). Both disagreements are on cells with $|\Delta_{\text{cov}}|$ at or below the noise floor (NeurIPS Black-in-AI panel: GPT 45 $\rightarrow$ 44 vs Claude 47 $\rightarrow$ 47; arXiv Megagauss: GPT 33 $\rightarrow$ 34 vs Claude 39 $\rightarrow$ 37 — judges differ in absolute counts by 4–6 but stay within ± 2 on cascade direction). The 10/10 coverage agreement on the original 10 cells degrades to 20/22 only on near-tie cells. Crucially, on the two largest-cov-gain cells in the new set (Harvard Moot Court +50pp, Doudna CRISPR +24pp), both judges concur.

Hallucination disagreements (7/22). The new-cell disagreements split into two patterns. (i) *Near-tie cells* where a ± 1 judge difference flips direction: NeurIPS panel (6h baseline), arXiv Fairness AI (14h $\rightarrow$ 15h vs 11h $\rightarrow$ 9h). (ii) *High-count cells* where the two judges differ in absolute counts by

10+ but the cascade-direction sign also flips: SP-5100 (47 $\rightarrow$ 45 on GPT-5.4 vs 36 $\rightarrow$ 68 on Claude), arXiv Megagauss (36 $\rightarrow$ 32 vs 19 $\rightarrow$ 13). Claude as judge systematically reports fewer hallucinations than GPT-5.4 (median Claude / GPT-5.4 hallucination ratio = 0.71 across all 41 Gemini condition-cell pairs with GPT-5.4 > 0); on cells where the absolute count is high, the proportional disagreement can flip cascade direction even when both judges agree that “many claims are unsupported.” The per-cell hallucination magnitude finding is therefore judge-sensitive on the new long-paper cells; the cross-vendor Mode B claim (Claude C_1 produces 17–47 unsupported expansions) should be read as “GPT-5.4 sees this many; Claude judges Claude’s own C_1 at 22 (Fairness AI), 13 (Megagauss), and 36 (SP-5100) — still consistent with the bibliographic-expansion signature, at lower absolute counts.”

Direction-robust findings. The key qualitative findings survive both judges: (a) the Mode B-driven counter-cells (Natural Vibration, Fairness AI cov, Perovskite cov) all show $C_1 < C_0$ on coverage under both judges; (b) the strong cascade-positive cells (Karpathy review, MIT 6.034, Harvard Moot Court, 3B1B, Doudna, Stanford Decarb, Sapolsky, Thermal) show $C_1 > C_0$ on both axes under both judges; (c) the inverse correlation between C_0 baseline and Δ_{cov} holds under both judges ($r = -0.48$ on Claude-scored data vs. -0.45 on GPT-5.4, $n = 21$ both).

Full per-cell tables for the original 10 cells in Appendix B; per-cell deltas for the new 12 cells are saved in `research_log/inter_judge_summary_new12.json`.

4 Discussion

Cascade as a deployment choice. For long-form multimodal review generation the right cascade choice is governed by source length, citation density, and the model’s relative text-vs-multimodal reasoning strength. We give a small predictive checklist (Table 5) drawn from the cells in this paper.

Table 5: Practical checklist drawn from this paper’s cells. “Recommended action” is the cascade variant with the best Pareto position on the corresponding cell.

Source / model class	Diagnosis	Recommended action
Audio talk, single speaker, sub-hour	Cascade strongly Pareto-positive on capable models.	Default C_1 .
Multi-speaker meeting (60 min)	Cascade still Pareto-positive (SNAC: +24.5 pp cov).	Default C_1 .
Moderate paper (40–70 pp)	Cascade Pareto-positive across vendors.	Default C_1 .
Long paper, low C_0 baseline, moderate citation density (Gemini Heat Pipes)	Mode A dominant.	C_{1c}^{concat} (chunked, no merge).
Long paper, low C_0 baseline, plus signs of Mode B (Mimo Heat Pipes)	Modes A and B both observed; no isolation experiment.	C_{1c}^{concat} for coverage; consider C_1^{strip} as a parallel control. Validate per source.
Long paper, citation-heavy, well-known refs (Claude Heat Pipes)	Mode B dominant.	Use C_0 or C_1^{strip} ; do not default to C_1 .

Limitations. *Cross-vendor n.* Cross-vendor cascade evidence on long papers is now strongest on four papers tested with Claude C_1 (Heat Pipes, SP-5100, Fairness AI, Megagauss) and one with Claude $C_0 + C_1$ (Thermal). Claude C_0 on the longest papers exceeded request-size limits even with the 1M-context beta, so the cleanest end-to-end vs cascade comparison on Claude is bottlenecked by Anthropic’s per-request payload cap rather than by model context. *Statistical noise.* We report bootstrap 95% CIs and sign tests for aggregate effects across cells (§3.1) and Wilson 95% CIs on per-cell coverage rates (Appendix C). The per-cell “within noise” label ($|\Delta h| \leq 1$ on $h < 10$, $|\Delta_{\text{cov}}| \leq 0.04$) is a presentation convention calibrated against inter-judge direction flips, not a measured Poisson dispersion. *Judge variance.* The inter-judge replication now covers all 21 Gemini cells but does not address single-judge intra-rater variance from re-runs with different random seeds. We did not characterize GPT-5.4’s seed sensitivity directly. The hallucination-count disagreements on high-count cells (§3.7) reveal a systematic per-judge bias: Claude reports $\sim 0.71 \times$ fewer unsupported claims than GPT-5.4 across the same condition-cell pairs.

Claude Pass-2 and GPT-5.4 judge stochasticity. A two-run replication on three Claude cells (Heat Pipes, SP-5100, Fairness AI) plus 3–4 re-judgments of two single Claude review files reveals two distinct stochasticity sources: (a) Claude Opus 4.7 produces materially different reviews on identical inputs — hallucination counts swing $|\Delta h|$ up to 73 between two calls (Table 4); Claude Opus 4.7 deprecates the `temperature` parameter so determinism cannot be enforced from the SDK. (b) GPT-5.4 with `reasoning_effort=high` is essentially deterministic on low-count cells (± 1 across three re-judgments of the same Heat Pipes Claude review) but substantially stochastic on high-count cells (± 25 across four re-judgments of the same SP-5100 Claude review); we attribute this to the long unsupported-claims list bumping against our 24k-token completion budget. Coverage is much more stable than hallucination counts under both sources. The Mode B *observation* is robust to both noise sources (Claude C_1 hallucinates many citations across all runs and judge calls); the Mode B *fix* is not (§3.5). *Probe and reference bias.* The reference transcript is produced by a single vendor (Gemini 3.1 Pro for both OCR and ASR), and the probes are extracted by the same model that scores them (GPT-5.4 for both, with judge calls routed via OpenRouter due to scope restrictions on the local OpenAI key). Two distinct biases follow: (1) *reference bias* — a fact dropped by Gemini’s reference is invisible to the protocol regardless of whether C_0 or C_1 reproduces it, advantaging models that share Gemini’s transcription priors; (2) *probe-circularity bias* — the same model decides both “what counts as a probe-worthy fact” and “does the review cover it,” so any systematic blind spot in extraction propagates into scoring undetected. The natural next-step ablations — (a) re-extract probes with Claude on a subset of cells and re-score, and (b) substitute a non-Gemini reference transcript on one cell — are open. *Mode-B isolation.* The citation-stripping intervention was tested only on the original Heat Pipes Claude and Natural Vibration Gemini cells; we did not run it on the three new long-paper Mode B cells (SP-5100, Fairness AI, Megagauss), so the *intervention* side of Mode B is still single-cell. The Mode B *observation* now has $n = 4$ Claude cells but the matched *fix* only $n = 2$ (one positive, one negative). *Mode-A intervention scope.* Chunked Pass-2 was tested only on Heat Pipes; we did not run it on the three arXiv cells where Mode A signatures appear most strongly. *Mimo audio bitrate.* Mimo cells ran at 16kbps mono mp3 (12kbps for the 85-min Sapolsky cell) to fit OpenRouter’s 10MB base64 audio input limit; the Gemini reference used 32kbps. Some Mimo audio cells therefore conflate audio-modality-specific weakness in Mimo with audio quality degradation. *Claude/Mimo long-paper C_0 .* On the four longest paper cells (SP-5100 176pp and the three arXiv surveys 53–80pp at 200 DPI full color), Claude C_0 exceeds Anthropic’s per-request payload limit even with the 1M-context beta, and Mimo’s image-multimodal endpoint returns empty/no review. Those cells contribute Claude C_1 / Mimo C_1 numbers only, not the within-vendor $C_0 \rightarrow C_1$ direction. We did not down-sample the renderings to a smaller resolution to fit the Claude payload cap in this revision. *Task scope.* All measurements are on long-form review or meeting-minutes generation. We make no claim that the same cascade direction holds on short-form QA, structured extraction, code generation, or agentic tool use; Phase-1 results from the same project (released alongside this paper) explicitly found end-to-end $\geq$ cascade on short-form MCQ benchmarks (MMAR, DocVQA, ChartQA) for current-generation frontier models. *Other vendor API constraints.* Claude does not natively accept audio multimodal input; gpt-audio refuses text-only Pass-2 with HTTP 400. We could not realize the same-weights cascade on those models.

Open questions. (i) Multi-seed Claude Pass-2 averaging on ≥ 5 independent calls per (cell, condition) pair, to characterize the stochasticity envelope and re-evaluate whether citation stripping has any signed effect on Claude after averaging. (ii) Whether Mode B generalizes beyond bibliographic surface forms to other “famous abbreviations” (model names, equation labels, dataset references). (iii) An iterative-refinement cascade variant: have Pass-2 emit per-claim citations back to spans of the Pass-1 transcript, then run a faithfulness filter that rejects claims whose cited span does not support them. (iv) Whether re-stripping citations after sub-summarization (§3.5) recovers additivity of the two fixes.

5 Related work

Speech. Cuervo et al. [1] document a text/speech understanding gap. The Cascade Equivalence Hypothesis [2] shows speech-LLMs internally behave like ASR $\rightarrow$ LLM cascades and that external cascades dominate at low SNR. X-Talk [3] argues optimized cascaded speech-to-speech pipelines are competitive on latency. Step-Audio-R1 [4] reports default audio-LLMs perform better with minimal reasoning and proposes modality-grounded reasoning distillation; we read this as consistent with our

Mode B finding under a different name (“audio reasoning needs to be grounded in audio, not text”) — structurally analogous to “Pass-2 from text leaks training prior”.

Documents. DocVLM [5] reports OCR-augmented VLMs beating pure-vision VLMs at fixed token budget on DocVQA. The OCR-Reasoning Benchmark [6] shows pure-OCR cascades cap below 50% on text-rich reasoning where spatial information matters. ColPali [7] shows vision-only retrieval beats OCR for retrieval (a different task class).

Cascade theory. Bengio’s Consciousness Prior [10] argues for sparse discrete bottlenecks between perception and reasoning. Our externalized text bottleneck is a deployment-time analogue. Discrete JEPA [11] makes a related architectural argument within vision.

Failure-mode taxonomies. The closest precedent for our two-mode taxonomy is the cascade-vs-end-to-end work on document understanding [5] and on speech [2], both of which characterize *when* cascades win or lose without naming the mechanisms. To our knowledge this paper is the first to propose a vendor-conditional Pareto frontier with named mechanisms (Mode A, Mode B) and matched engineering interventions.

Citation-grounded generation and faithfulness. Mode B — the model fills in citation surface forms from training-time knowledge of the referenced literature — is a self-cascade analogue of the unfaithfulness problem that motivates retrieval-augmented generation [12], post-hoc attribution and revision [13], and faithful-attribution evaluations such as Attribution to Identified Sources [14]. Where those lines of work address ungrounded generation by retrieving and attributing external evidence, Mode B isolates the same failure inside a closed self-cascade in which the “evidence” is the model’s own Pass-1 transcript: the model has the document in front of it and still embellishes the abbreviation. Our citation-stripping intervention is a source-anonymization defense applied at the cascade boundary; the finding that stripping has *vendor-conditional* sign (helping Claude, hurting Gemini) suggests this class of defense is not universally safe and needs to be tuned per model.

6 Conclusion

The cascade gap on long-form multimodal review is real at $n = 21$: mean $\Delta_{\text{halluc}} = -3.86$ ($p = 0.0015$), mean $\Delta_{\text{cov}} = +0.118$ ($p = 0.027$), inverse correlation $r = -0.45$ with bootstrap CI $[-0.73, -0.10]$. But the gain is vendor- and source-conditional. Two failure modes account for the cells where the cascade fails to dominate end-to-end: a compression bottleneck (now observed on Heat Pipes Gemini and two of three modern arXiv surveys on Gemini) and a training-prior leak (now observed on Claude across four long citation-heavy papers). Two engineering interventions evaluated under replication: chunked Pass-2 replicates cleanly across three long-paper Gemini cells (C_{1c}^{concat} consistently lifts coverage at a hallucination cost); citation stripping does not replicate cleanly on Claude (the apparent strip-helps-Claude effect on Heat Pipes is within the Pass-2 noise band measured at two runs). The cascade is therefore not a universal architecture but a deployment-time tool with a small set of predictable failure modes and matched fixes. A leave-one-out predictor on simple cell features achieves AUC 0.65 at $n = 21$; the residual error sits on Mode B-prone cells, suggesting that a citation-surface-form feature would improve it further. Practitioners should pick the cascade variant by source length, citation density, page-rendering density, and model-text-reasoning strength, not as a default.

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A Protocol details

Reference quality. We use Gemini 3.1 Pro for OCR/ASR with chunking. Single-call OCR on a 104-page paper produced a degenerate TOC dot-leader loop (113KB output, 56k “words” but only 264 real-word tokens, 6 page markers out of 104). Chunking (1 page per call, max 8192 tokens per call output) eliminates this failure mode. For audio, we chunk into 30-min segments with 5-second overlap. We discovered a silent server-side truncation on one audio chunk (chunk 1 of IETF SNAC produced 1981 words instead of the expected ≈ 4368 , ending mid-sentence with “Uh, well, see, yeah, that’ ”). The protocol grew an automatic chunk-level word-count parity check between reference and C_1 Pass-1 after this incident.

Sources. Audio: Karpathy “State of GPT” 42-min, IETF SNAC WG interim 2026-04-16 (60 min, ≈ 5 speakers, recorded post the most likely training-data cutoff for the foundation models in the loop). Papers: NASA Tech Memos sourced from NASA NTRS, IDs 19720009221 (9pp), 19680000724 (18pp), 19690008405 (41pp), 19690013408 (42pp), 19700023812 (66pp), 19700025120 (104pp). All papers are from 1968–1972 and rendered as page images at 200 DPI.

Models and snapshots. Gemini 3.1 Pro Preview (gemini-3.1-pro-preview); Claude Opus 4.7 (claude-opus-4-7); Mimo v2-omni (xiaomi/mimo-v2-omni via OpenRouter); judge GPT-5.4 (reasoning_effort=high).

Mimo audio constraint. OpenRouter’s Mimo provider enforces a 10MB base64 audio input limit. Karpathy required re-encoding from 32kbps to 16kbps mono mp3 to fit; SNAC at 60 min could not fit even at 16kbps. The Mimo audio Karpathy result is at 16kbps; the corresponding Gemini C_0 baseline on the same source used 32kbps. Mimo audio Karpathy numbers: $C_0 = 4$ unsupported claims, coverage $29/51 = 0.569$, 1509 review words; $C_1 = 4$ unsupported claims, coverage $30/51 = 0.588$, 1268 review words. Both axes are within noise across the $C_0 \rightarrow C_1$ transition, so we treat the cell as a cascade wash rather than as a positive or negative cross-vendor result.

gpt-audio. OpenAI gpt-audio (via OpenRouter) accepted audio input cleanly but refused text-only Pass-2 with HTTP 400 “This model requires that either input content or output modality contain audio”. Same-weights cascade is therefore not realizable on this model. The C_0 alone scored 10 hallucinations / 0.765 coverage on Karpathy.

B Inter-judge details

Tables 6 and 7 summarize per-cell GPT-5.4 vs Claude judging on the two axes separately. Coverage counts are out of n probes per cell, listed in the n column.

Table 6: Inter-judge hallucination counts. Direction $\checkmark$ means both judges agree on the sign of $C_0 \rightarrow C_1$.

Cell	GPT-5.4		Claude		Direction agree?
	C_0	C_1	C_0	C_1	
Karpathy review	13	3	7	4	$\checkmark$
Karpathy mtg-min	4	2	1	0	$\checkmark$
SNAC mtg-min	14	11	10	11	$\times$
Wagging Tail (9pp)	3	1	2	2	$\times$
Env Test (18pp)	5	4	0	2	$\times$
Dynamic Response (41pp)	3	2	1	0	$\checkmark$
Natural Vibration (42pp)	4	5	2	4	$\checkmark$
Thermal Analysis (66pp)	5	2	2	1	$\checkmark$
Heat Pipes (104pp)	14	11	12	11	$\checkmark$
Thermal Analysis Claude	11	4	9	4	$\checkmark$

Table 7: Inter-judge probe-coverage counts (probes covered out of n). Direction $\checkmark$ means both judges agree on the sign of $C_0 \rightarrow C_1$.

Cell	n	GPT-5.4		Claude		Direction agree?
		C_0	C_1	C_0	C_1	
Karpathy review	51	32	49	38	49	$\checkmark$
Karpathy mtg-min	50	8	17	13	21	$\checkmark$
SNAC mtg-min	49	14	26	15	28	$\checkmark$
Wagging Tail (9pp)	49	39	38	40	38	$\checkmark$
Env Test (18pp)	50	44	46	46	49	$\checkmark$
Dynamic Response (41pp)	50	29	35	32	36	$\checkmark$
Natural Vibration (42pp)	47	31	23	32	28	$\checkmark$
Thermal Analysis (66pp)	50	34	42	34	45	$\checkmark$
Heat Pipes (104pp)	50	36	39	39	41	$\checkmark$
Thermal Analysis Claude	50	46	50	46	49	$\checkmark$

Per-probe agreement (the fraction of probes the two judges classify identically as COVERED / MISSING) ranges from 0.882 to 1.000 across the (cell, condition) pairs, with median ≈ 0.95 . Two of the 3 hallucination-direction disagreements (Wagging Tail, Env Test) are on cells with ≤ 5 absolute

hallucinations under both judges — the near-tie regime where a ± 1 judge difference flips direction (consistent with our $|\Delta h| \leq 1$ noise floor from §2). The third (SNAC) is not a near-tie cell: GPT-5.4 reports $14 \rightarrow 11$ (a clear decrease, outside the noise floor) while Claude reports $10 \rightarrow 11$ (a near-tie); the disagreement is driven by the two judges assigning different C_0 baselines (14 vs 10) while agreeing on C_1 (both 11). Coverage direction is unanimous across all 10 cells.

C All 21 cells: raw counts and Wilson 95% CIs

Table 8 gives the raw C_0 vs C_1 counts for every cell, with Wilson 95% CIs on the coverage rates. The Δ columns are point estimates; CIs on coverage and Wilson reflect the per-cell uncertainty before considering judge variance.

Table 8: All 21 Gemini base cells. Halluc. is the raw unsupported-claim count. Cov. is k/n (covered probes out of n) with Wilson 95% CI. Cells where C_1 *does not* strictly improve on at least one axis above the noise floor are marked with a $\dagger$ in the Δ column.

Cell	len	C_0 halluc	C_0 cov [95% CI]	C_1 halluc	C_1 cov [95% CI]	Δ_{cov}
<i>Audio (11 cells)</i>						
Karpathy review	42m	13	0.627 [0.490, 0.747]	3	0.961 [0.868, 0.989]	+0.333
Karpathy mtg-min	42m	4	0.160 [0.083, 0.285]	2	0.340 [0.224, 0.478]	+0.180
SNAC mtg-min	60m	14	0.286 [0.178, 0.424]	11	0.531 [0.394, 0.663]	+0.245
3B1B Attention	26m	11	0.880 [0.762, 0.944]	0	0.960 [0.864, 0.989]	+0.080
Veritasium Math	32m	14	0.820 [0.692, 0.903]	6	0.960 [0.864, 0.989]	+0.140
MIT 6.034 Winston	47m	17	0.771 [0.633, 0.866]	4	0.979 [0.890, 0.996]	+0.208
NeurIPS BinAI panel	31m	6	0.918 [0.803, 0.969]	4	0.898 [0.778, 0.957]	−0.020 $\dagger$
Doudna CRISPR	48m	6	0.680 [0.542, 0.792]	3	0.920 [0.812, 0.968]	+0.240
Stanford Decarb	72m	11	0.700 [0.562, 0.809]	6	0.900 [0.786, 0.957]	+0.200
Harvard Moot Court	82m	11	0.360 [0.241, 0.499]	4	0.860 [0.736, 0.932]	+0.500
Sapolsky Behavioral	85m	19	0.592 [0.450, 0.720]	15	0.939 [0.834, 0.980]	+0.347
<i>Paper, NASA NTRS 1968–1972</i>						
Wagging Tail	9p	3	0.796 [0.664, 0.885]	1	0.776 [0.641, 0.870]	−0.020 $\dagger$
Env Test	18p	5	0.880 [0.762, 0.944]	4	0.920 [0.812, 0.968]	+0.040 $\dagger$
Dynamic Response	41p	3	0.580 [0.442, 0.706]	2	0.700 [0.562, 0.809]	+0.120
Natural Vibration	42p	4	0.660 [0.517, 0.778]	5	0.489 [0.353, 0.628]	−0.170 $\dagger$
Thermal Analysis	66p	5	0.680 [0.542, 0.792]	2	0.840 [0.715, 0.917]	+0.160
Heat Pipes	104p	14	0.720 [0.583, 0.825]	11	0.780 [0.648, 0.872]	+0.060
SP-5100 Shock	176p	47	0.521 [0.383, 0.655]	45	0.604 [0.463, 0.730]	+0.083
<i>Paper, arXiv 2026 (post-cutoff)</i>						
arXiv Fairness AI	53p	14	0.500 [0.366, 0.634]	15	0.400 [0.276, 0.538]	−0.100 $\dagger$
arXiv Megagauss	75p	36	0.660 [0.522, 0.776]	32	0.680 [0.542, 0.792]	+0.020 $\dagger$
arXiv Perovskite	80p	2	0.900 [0.786, 0.957]	3	0.740 [0.604, 0.841]	−0.160 $\dagger$

D Per-cell hallucinations per 1k output words

Table 9: Hallucination density (per 1k output words) on Thermal Analysis 66pp.

	C_0	C_1
Gemini 3.1 Pro	4.2	1.4
Claude Opus 4.7	3.7	1.4
Mimo v2-omni	4.9	5.4

End-to-end hallucination rates differ per vendor (3.7 / 4.2 / 4.9 per 1k); cascade brings Gemini and Claude to a common ≈ 1.4 per 1k floor on the moderate-length paper. Mimo does not converge to this floor on the cascade.

Claude’s per-1k hallucination rate jumps from 1.2 (C_0) to 9.6 (C_1) on Heat Pipes — an $8\times$ increase that survives length normalization. The same model on the same protocol on Thermal 66pp moved $3.7 \rightarrow 1.4$ (cascade improves). Citation stripping pulls Heat Pipes Claude back to 5.3/1k, halving the cascade penalty. Gemini and Mimo on the same paper sit at 6–8/1k regardless of cascade variant, indicating that the 1.4/1k floor reached on Thermal is not a property of the cascade in general, but of the specific (model, source) interaction.

Table 10: Hallucination density (per 1k output words) on Heat Pipes 104pp. Claude C_1 at 9.6/1k is the largest value in our dataset and dwarfs Claude’s Thermal C_1 rate of 1.4/1k, supporting the Mode B interpretation as a length- and citation-density-triggered effect rather than a per-token verbosity effect.

	C_0	C_1
Gemini 3.1 Pro	7.6	6.1
Claude Opus 4.7	1.2	9.6
Claude Opus 4.7 (C_1^{strip})	—	5.3
Mimo v2-omni	7.0	6.9

E Specific hallucination examples (Heat Pipes Claude)

The 20 unsupported claims from Claude C_1 on Heat Pipes 104pp are dominated by training-prior expansions of bibliographic abbreviations:

- “Cotter’s 1965 analysis (LA-3246-MS)” — transcript cites only “Reference 2 (Cotter)”; year and report number are training-data details.
- “correlation from Allingham and McEntire (1961)” — transcript only says “Ref. 12”; author names and year are training-data details.
- “McAdams’ laminar horizontal-tube formula” — transcript says “Reference 13 for laminar condensation on horizontal tubes”; “McAdams” is a training-data detail.
- “Vapor chamber-fin radiator (Haller, Lieblein, and Lindow at NASA Lewis)” — transcript says only “proposed by Haller, et al of NASA”.

In addition, one outright misreading: the transcript states the no-boiling condition as $p_v - p_\ell \leq 2\sigma \cos \theta / r_c$; Claude’s review writes $\geq$. This is a sign error in Pass-2 reading of Pass-1 text, not training-prior leak.

F Code and data availability

All code, prompts, and per-cell judgments are released in the project repository. The pilot pipelines (pilot/pilot_audio_review.py, pilot/pilot_paper_review.py, pilot/pilot_paper_review_claude.py, pilot/pilot_paper_review_mimo.py) and post-processing scripts (pilot/chunked_pass2.py, pilot/citation_strip.py, pilot/inter_judge_claude.py) reproduce every cell in the paper.